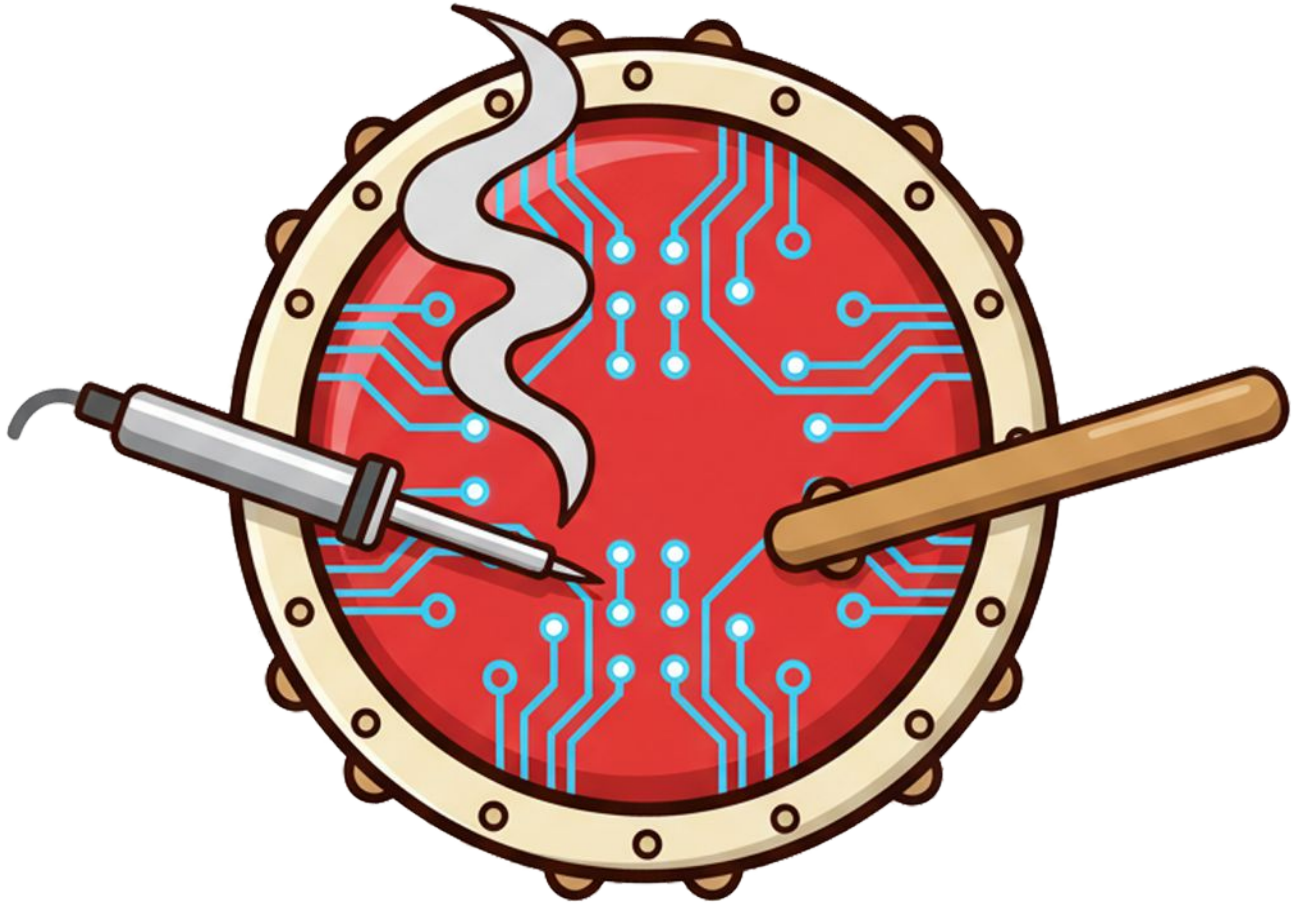




Ouchi**Taiko** Project

By KillerQ

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**In Japanese, 'ouchi' (おうち) means 'home' and 'taiko' (太鼓) means 'drum.'
Together, 'OuchiTaiko' represents the joy of bringing the authentic
Taiko experience from the Arcade into your own space.**

1: Project Overview



Congratulations! Welcome to "KillerQ's OuchiTaiko Project: Arcade Controller Build Guide." You're about to build the most functional and comprehensive arcade-scale home Taiko drum controller ever designed. This open-source guide represents 8 months of research and development, bringing the authentic arcade experience home without the high cost.

The OuchiTaiko Project features a pioneering adaptive drum design—the first of its kind for any home arcade controller—addressing the limited availability and high cost of commercial units.

Smarter. Simpler. Better.

Instead of complex analog circuits with operational amplifiers and custom PCBs, this design uses adaptive software intelligence to achieve superior performance through elegant simplicity.

All you need is enthusiasm and basic soldering and woodworking skills. The components linked in this guide create a perfectly functional controller when used together. Feel free to substitute compatible alternatives if you find cheaper options that achieve the same results.

What You'll Build: Next-Generation Features

You will construct a professional arcade-scale Taiko drum controller engineered for premium performance and maximum compatibility, featuring:

Unprecedented Performance & Design:

- **4 Velocity-Sensitive Zones:** Enhanced mechanical and electronic false-trigger isolation
- **Custom Arcade Sensors:** Drum trigger sensors with custom-designed suspension mounting that precisely mimics Japanese arcade machines (a public first-time release)
- **Adaptive Baseline Technology:** Pioneering velocity-sensing for authentic Big Note scoring and automatic environment adjustment (a first for any home arcade controller)
- **Zero Coding Required:** Drag-and-drop firmware for simple setup

Advanced Hardware & Connectivity:

- **OLED Display:** On-the-fly mode selection, settings, calibration, and real-time hit feedback

IDLE SCREEN SHOWS:

- Controller mode (top)
- Streak counter (live roll count)
- 4 animated characters (react to hits)
- Player LEDs (PS4 mode, top-right)
- Menu hint (bottom)

VISUAL FEEDBACK:

- Each character = one drum pad (left to right: Ka L, Don L, Don R, Ka R)
- Characters squish when hit
- 60fps smooth animation
- Instant hit confirmation

HARDWARE:

- **14 Navigation Buttons:** Full in-game navigation regardless of game version
- **Built-in 3.5mm Headphone Jack:** Lag-free audio right at the drum
- **Professional Mounting:** Optimal hardware stabilization via adjustable, angled speaker stand
- **Performance:** Rivals high-end manufactured controllers and commercial Arcade units

Controller Input Modes for Maximum Compatibility:

1. HORI NSW-079 Taiko Drum (Switch)
2. HORI PS4-095 Taiko Drum (PS4, with time limitation)
3. Switch Pro Controller
4. XInput & XInput Analog (Compatible with TaikoArcadeLoader)
5. DualShock 4 (PC/Steam only)
6. DualShock 3
7. Keyboard Player 1
8. Keyboard Player 2
9. Analog Player 1
10. Analog Player 2
11. MIDI
12. Debug

Demo videos:

- Finished Drum: Link [here](#)
- Some Gameplay: Link [here](#)
- Roll Polling Rate Demo: Link [here](#)
- Hit Velocity Detection: Link [here](#)

2: Parts List for Electronics

The Amazon links reference parts that are 100% compatible when used together. You can source these items from any supplier as long as they meet the specifications.

Important: "Quantity" refers to the number of individual items you need, NOT the package size sold at the link. For example, this project requires 4 individual 27mm piezo sensors, but the Amazon link sells a 20-pack.

	Item	Quantity	Product Link
1	XIAO RP2040 board (Seeed Studio)	1	https://a.co/d/g9dFG88
2	MCP23017 I2C expander Board	1	https://a.co/d/er7ux8r
3	1N4148 diodes	4	https://a.co/d/jlNKdaJ
4	100nF capacitor (same as 0.1 uf)	4	https://a.co/d/fP04dyW
5	3.5mm TRS FEMALE jacks w/ screw terminals	4	https://a.co/d/2umR9us
6	3.5mm TRS male w/ standard ends	4	https://a.co/d/414YG9z
7	27mm Piezo sensors	4	https://a.co/d/hbacbaF
8	128x64 Mono OLED display (I2C)	1	https://a.co/d/bHAtiY2
9	Tactile buttons (6mm or your choice)	14	https://a.co/d/2j6JNsp
10	22 AWG Siamese wire	1	https://a.co/d/0pGa2wH
11	Small USB-A hub (2+ ports)	1	https://a.co/d/4vWTHAC
12	USB Audio DAC with 3.5mm output (32-bit, 48kHz)	1	https://a.co/d/au4BLDH
13	6 inch 3.5mm TRS audio extension cable (Male to Female)	1	https://a.co/d/cesBnrm
14	USB-A to USB-C converter	1	https://a.co/d/bG4QJUF
15	Short USB-C to USB-A extension cable	1	https://a.co/d/d61CuLt
16	6 inch, coiled 3.5mm TRS audio extension cable (Male to Male)	4	https://a.co/d/cSKAJ7D

17	3.5mm TRS barrel coupler	4	https://a.co/d/fOiYb5s
18	JST-XH 2.54mm Pitch connector kit	1	https://a.co/d/11FXUll
19	22 AWG Connection Ribbon Wire	1	https://a.co/d/1D5mJ8Z
20	2.54mm pitch Header Pins	1	https://a.co/d/6D4ftKM
21	Female DuPont connector wires	1	https://a.co/d/a7tXZma
22	Blank Protoboard Circuit Boards	3	https://a.co/d/2yOqCkd
23	10ft USB-C to USB-C Cable	1	https://a.co/d/6F0Ic5R

3: Parts List for Hardware

The Amazon links reference parts that are 100% compatible when used together. You can source these items from any supplier as long as they meet the specifications.

Important: "Quantity" refers to the number of individual items you need, NOT the package size sold at the link. For example, this project requires 8 individual M6x8mm bolts, but the Amazon link sells a 100-pack.

Required Tools and Supplies:

This guide assumes you have access to the following:

1. Laser cutter or CNC machine (or other means to cut wood)
2. Soldering iron and solder
3. Wire strippers/cutters
4. Screwdrivers
5. Hot glue gun
6. Strong clamps
7. Drill
8. Router
9. Sandpaper
10. Utility knife
11. Rubber mallet

	Item	Quantity	Product Link
1	6mm Cabinet-Grade MDF thick Wood. 1/4 Inch thick will work as well. Amount needed dictated by the sizes shown in the provided SVG files here)	1	Call your local lumberyard, or try these: https://www.homedepot.com/p/1-4-in-x-2-ft-x-4-ft-Medium-Density-Fiberboard-1508104/202089069
2	Strong Wood Glue	1	https://a.co/d/1uKv6cR
3	M3x8mm bolts	8	https://a.co/d/52Q8UtD
4	M3x5mm threaded inserts	8	https://a.co/d/bKB6OpW
5	M6x10mm wood threaded inserts	14	https://a.co/d/jhY5rYA
6	M6x20mm nylon bolts	14	https://a.co/d/bPitQiX
7	M6 threaded 20mmx15mm rubber isolators. One side M6 Female, other side with M6x18mm bolt	14	https://a.co/d/9dWHezk

8	3D printer filament (PLA)	1	https://a.co/d/7cCSDtJ
9	Gel Superglue	1	https://a.co/d/dxU7lfw
10	Loctite thread adhesive - Medium	1	https://a.co/d/2W890aJ
11	2.2mm thick Scuba Knit Neoprene fabric (only 4"x4" total needed)	1	https://a.co/d/dfMhZ4k
12	Finger Knobs with pass-through M6 threads	18	https://a.co/d/hjL3QQP
13	Mini PA speaker stand (or a stand with a compliant interface that matches your mount below)	1	https://a.co/d/2YkmhPj
14	Adjustable Angle Speaker bracket (or a compliant interface that matches your stand above)	1	https://a.co/d/gQioU8i
15	Rubber Taiko Drum Cover (optional, but crucial for the Arcade-Feel. Any thin sound-dampening material with bounce should work (towel, mouse pad, etc.)	1	https://taiko.ac/products/rubber-drum-pad

4: Build the Circuit

( For a detailed video overview of the circuit building process, click [here](#).)

This guide states the core objective of each step and provides example approaches through photos. Multiple layouts and methods can work.

Critical: Use a multimeter to test continuity as you go. A two-second check can save hours of troubleshooting caused by a poor solder joint or loose wire.

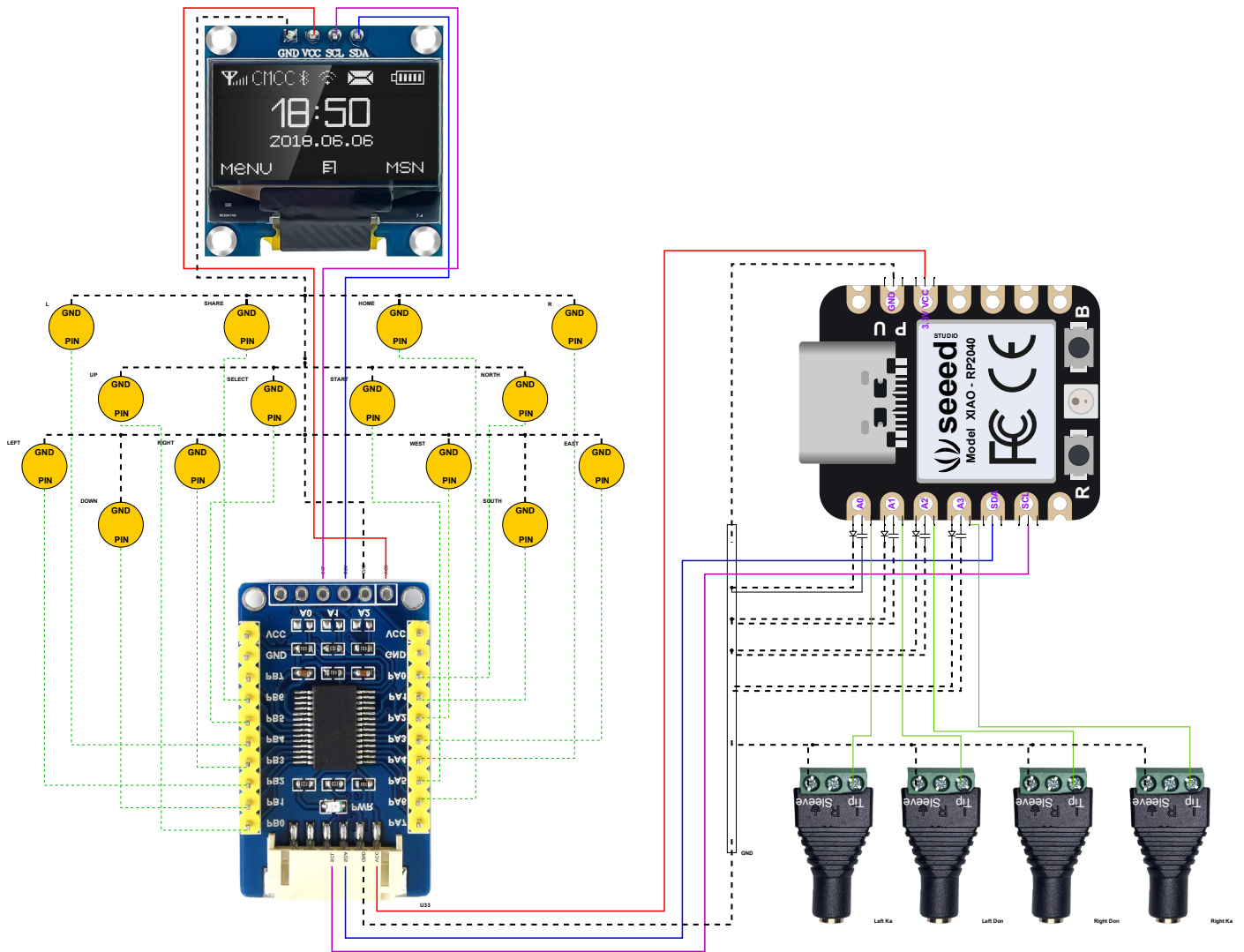
Review all steps, videos, and images before beginning to allow for optimal planning. This guide provides a 100% working system when followed as-is. However, as long as you adhere to the electrical theory and circuit structure, you can adjust and use creative freedom according to your comfort level.

Circuit Overview:

The Seeed Studio XIAO RP2040 board is the brain, power source, and host for the drum panel sensors. The external MCP23017 board handles the OLED display and navigation buttons.

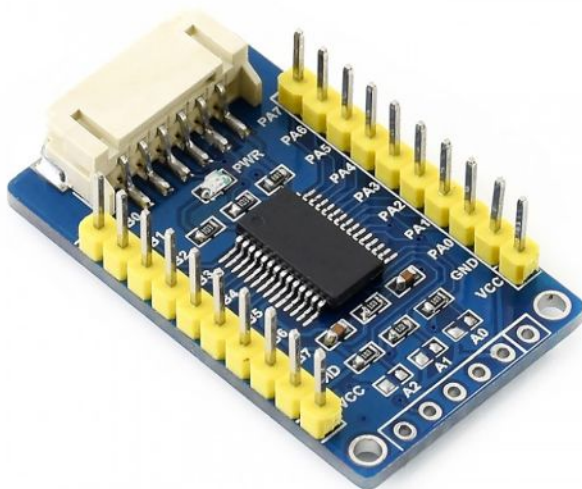
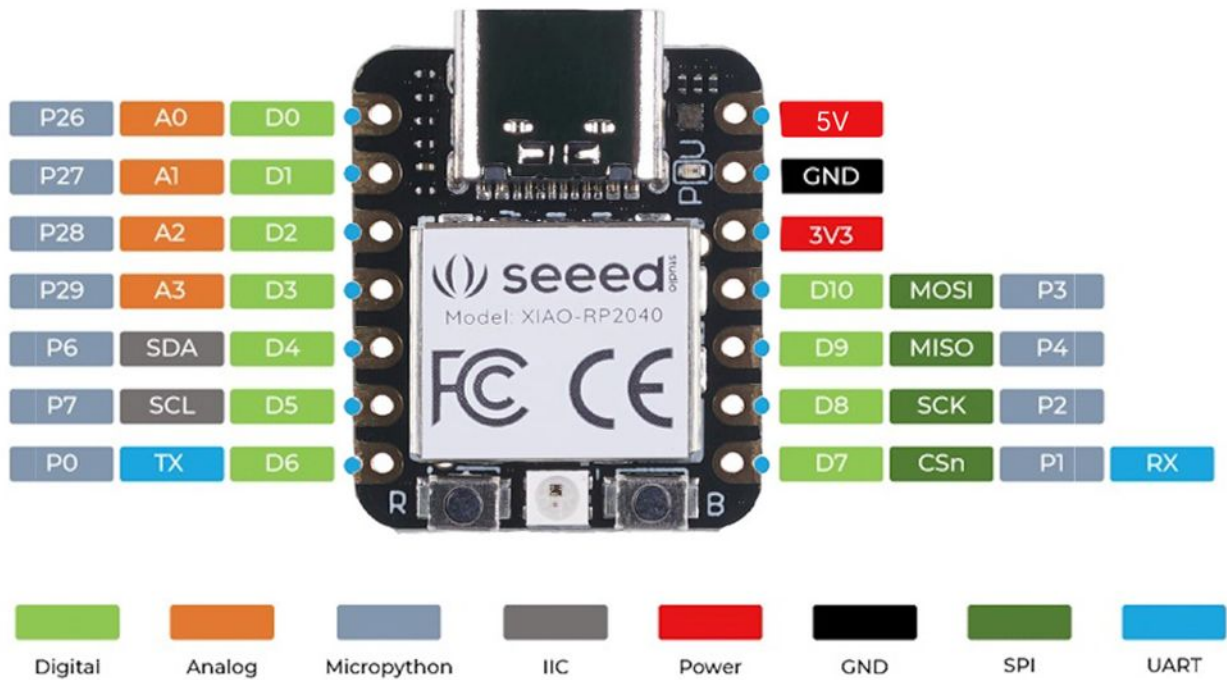
Tips for Success:

- Use color-coded wires for easy identification
- Test each phase with a multimeter for continuity before proceeding
- Label everything during assembly
- Take time with soldering connections

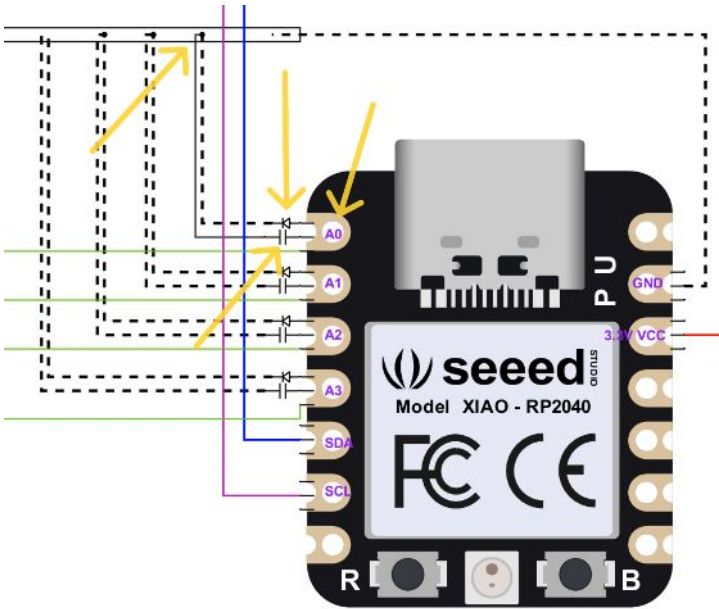


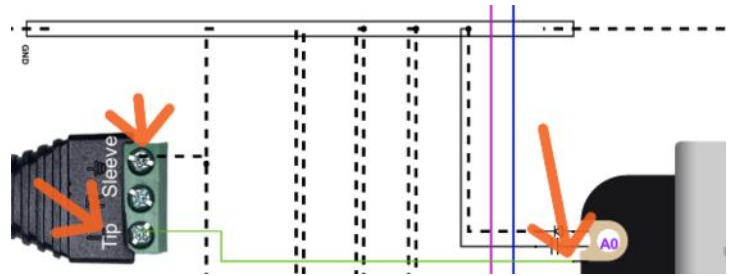
Note: The MCP23017 board in this diagram is flipped horizontally to show it mounted on the opposite side of the circuit board from the buttons. All connections are labeled correctly, just mirrored.

( Click [here](#) to download a full-resolution version of the Circuit Diagram for reference.)



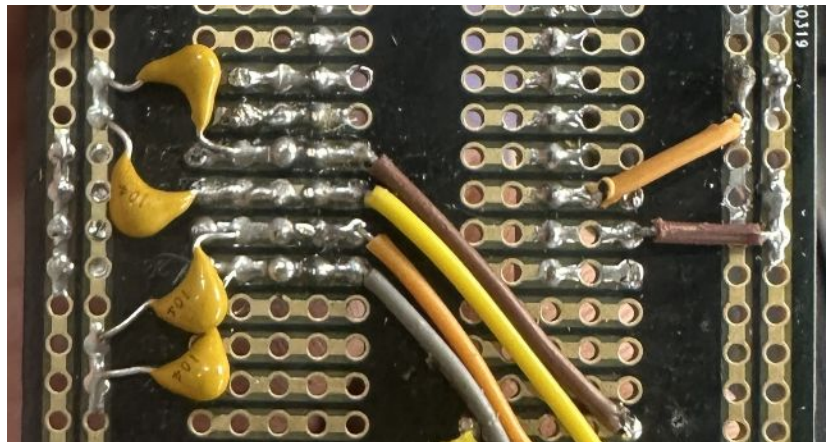
(Enlarged Pin-outs of the main boards for easy reference)

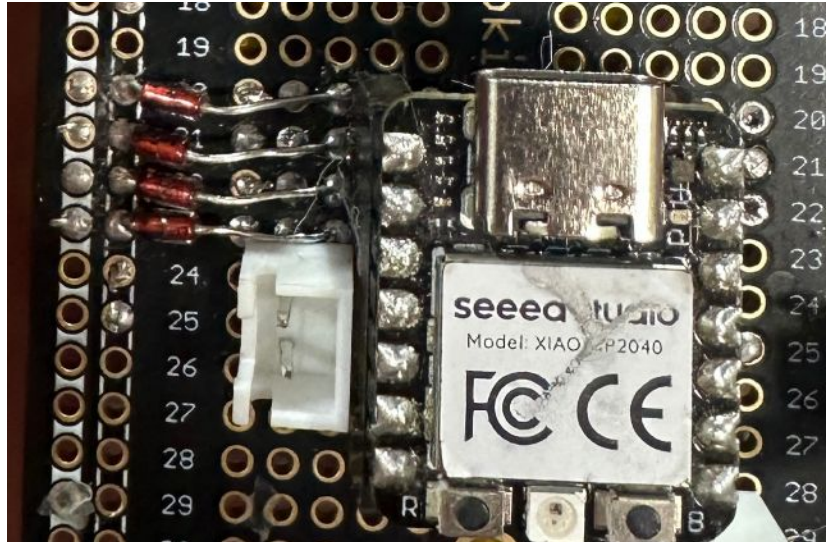
Step	Notes
A: XIAO Board Trigger Circuit	
(📄) The full-resolution circuit schematic can be downloaded reference here) A1.: Mount the XIAO RP2040 to a blank circuit board.	You will be soldering multiple components to the same pins, leaving ample space and utilization of the shared rows will help 
A2.: Connect the GND pin on the XIAO RP2040 to the common ground rail of the protoboard.	Be sure that the GND rails on both sides of the board are connected to each other. Some boards require a solder bridge to achieve this.
A3.: Connect the VCC 3.3V pin on the XIAO RP2040 to the common VCC Power rail of the protoboard.	Be sure that the VCC rails on both sides of the board are connected to each other. Some boards require a solder bridge to achieve this.
A4: Pin “P26” (Left Ka): 1) Solder the non-striped (anode) end of a 1N4148 diode to pin A0/P26. 2) Solder the other, striped end (cathode) of the diode to the common ground rail as close to the pin as possible. 3) Solder one end of a 100nF capacitor to pin A0/P26 (no polarity here, either end is fine). 4) Solder the other end of the 100nF capacitor to the common ground rail as close to the pin as possible. 5) Connect a yellow signal wire from pin A0/P26 to the <u>TIP</u> terminal of TRS Jack 1 (Left Ka) 6) Connect a black GND wire between the common ground rail and the SLEEVE terminal of TRS Jack 1.	



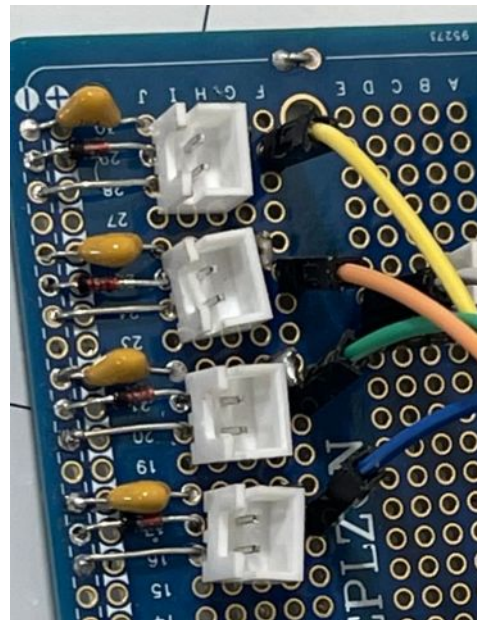
The Below photos show several variants of connecting the capacitors and diodes.

This example has the diodes on top of the board and the capacitors on the underside to allow more room to work.





This image puts both the capacitor and diode on the top side of the board.
Side note: this example also uses JST connectors to allow for modular connections of the TRS input jacks, so they are easily removable from the board (optional):



This is an example of a removable TRS input jack using a JST connection.

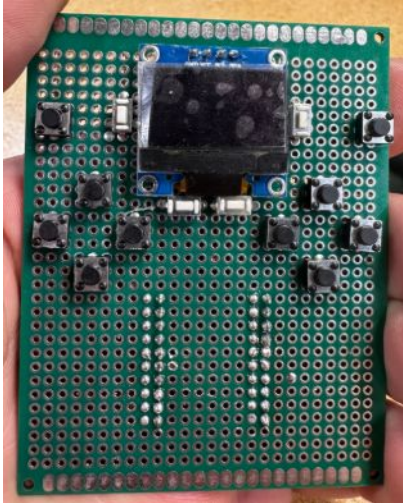


****Repeat the exact same 6-step process from A4 above for each of the three remaining analog pins. Use the following pin assignments and wire colors:**

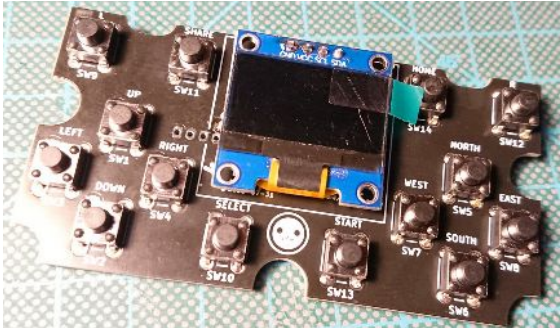
Step	Pin	Drum Zone	Signal Wire Color	TRS Jack #
A5.	A1/P27	Left Don	Orange	Jack 2
A6.	A2/P28	Right Don	Green	Jack 3
A7.	A3/P29	Right Ka	Blue	Jack 4

Summary: For each pin, complete these steps:

- 1. Solder 1N4148 diode (anode to pin, cathode to ground).**
- 2. Solder 100nF capacitor (one end to pin, other end to ground).**
- 3. Connect colored signal wire from pin to TRS Jack TIP terminal.**
- 4. Connect black wire from ground rail to TRS Jack SLEEVE terminal.**

I2C Control Board Circuit Mounting and Power	
A8. Mount the MCP23017 GPIO Expander on a blank board that is large enough to also house the OLED display and navigation buttons from steps later in this guide. Connect the wiring harness that came with the MCP23017 board. You can clip wires number 5 and 6 off. You will only be using the first 4 wires on this harness (red, black, blue, and yellow)	This board will house the display and buttons for the navigation. Plan enough space accordingly. In this photo, the MCP23017 is mounted on the rear of the board: 
A9. Using the wires from the harness you just plugged in, connect the VCC (red) wire to the VCC power rail on your main board that you built where the XIAO 2040 Board is mounted. Now, connect the GND (black) wire from the harness to the GND rail on your main board that you built where the XIAO 2040 Board is mounted.	This 2-wire connection will be powering the MCP23017 board and will also provide pass-through power for the OLED display
MCP23017 Data Connections	Note: Pins A7 and B7 will <u>not</u> be used for this project
A10. Connect the SDA (blue) wire on the MCP23017 harness to Pin 6 (SDA) on the XIAO 2040 board.	This is one of the two wires that sends the display and navigation buttons data back and forth during use.
A11. Connect the SCL (yellow) wire on the MCP23017 harness to Pin 7 (SCL) on the XIAO 2040 board.	This is the other of the two wires that sends the display and navigation buttons data back and forth during use.
I2C OLED Display Circuit	
A12. Mount the OLED display on the same board as the MCP23017 (and navigation buttons).	Reminder, this board will house the display and buttons for the gamepad navigation. Plan accordingly
A13. Connect a wire between the SDA soldering point on the MCP23017 board (opposite end of where you plugged the harness in) and the SDA pin on the OLED Display	

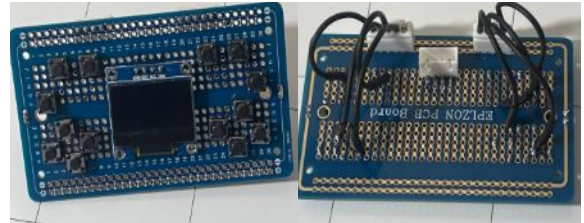
A14. Connect a wire between the SCL soldering point on the MCP23017 board (opposite end of where you plugged the harness in) and the SCL pin on the OLED Display	 The image shows two components: a blue MCP23017 I/O expander board and a small OLED display. On the MCP23017 board, the SCL pin is circled in yellow. On the OLED display, the SCL pin is also circled in yellow. This indicates the connection point for the SCL line.
A15. Connect a wire between the VCC soldering point on the MCP23017 board (opposite end of where you plugged the harness in) and the VCC pin on the OLED Display	 The image shows the same two components. On the MCP23017 board, the VCC pin is circled in yellow. On the OLED display, the VCC pin is also circled in yellow. This indicates the connection point for the VCC line.
A16. Connect a wire between the GND soldering point on the MCP23017 board (opposite end of where you plugged the harness in) and the GND pin on the OLED Display	 The image shows the same two components. On the MCP23017 board, the GND pin is circled in yellow. On the OLED display, the GND pin is also circled in yellow. This indicates the connection point for the GND line.

Navigation/Gamepad Buttons	
A17. Solder your 14 navigation buttons to your OLED circuit board in the same layout as the image to the right. This layout is similar to the traditional SWITCH Gamepad button and works best for this project.	 The image shows a custom PCB with 14 buttons arranged in a gamepad layout. The buttons are labeled: LEFT, UP, RIGHT, DOWN, SELECT, START, WEST, NORTH, EAST, SOUTH, and others. A small OLED display is mounted on the board. The layout is similar to a traditional SWITCH Gamepad button.

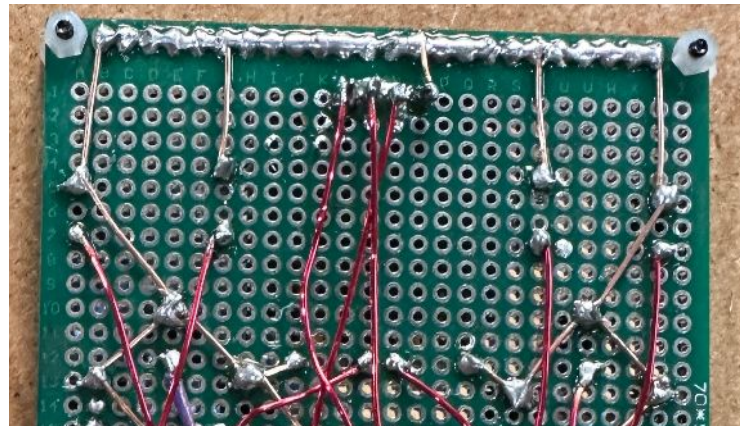
A18. Connect one leg of each of the 14 buttons button to the common ground rail.

You can also daisy-chain the ground wire between all buttons and then to a ground rail. Here are a few different example scenarios.

The first and second photos have all the grounds terminating at JST connectors that plug into a connector on the ground rail.



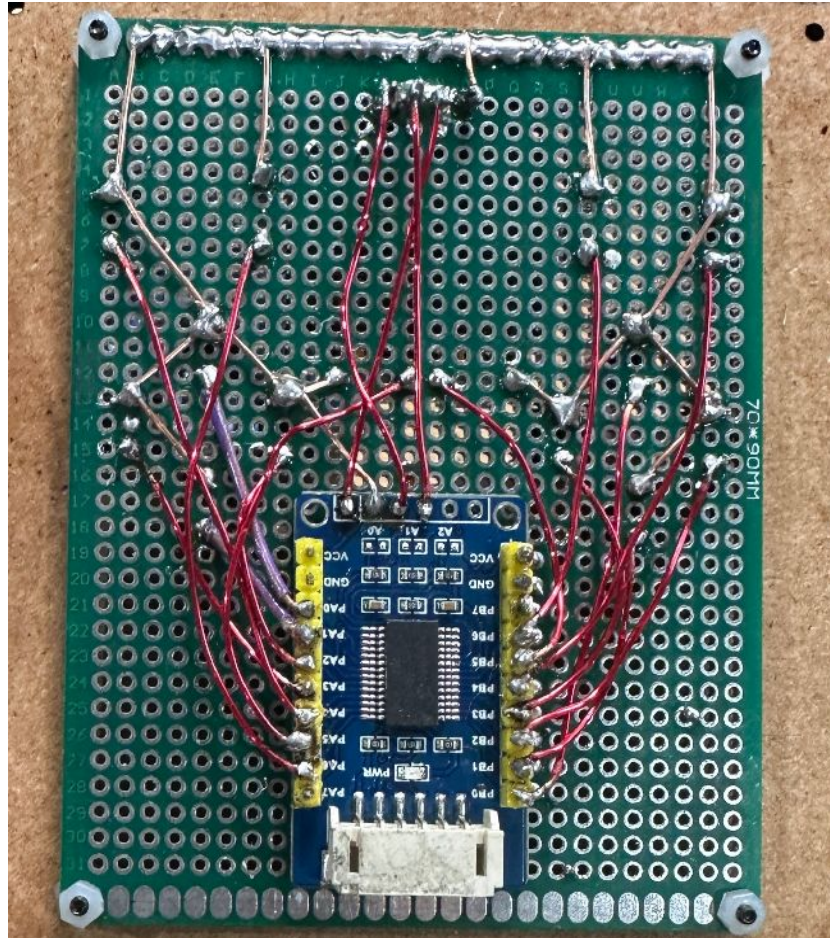
In the third photo, all of the buttons legs are oriented vertically. I chose to use the top leg as the ground connection. These top legs are all connected to a common ground rail at the top of the board. Some ground wires go straight to the ground rail; others connect to other wires first that eventually connect to the ground rail. All of these methods are sufficient.



A19. Next, connect the other leg of each button to the appropriate MCP23017 GPIO pins according to this guide (referencing the full circuit schematic may help here):

Button North: Connect to **A0**
Button South: Connect to **A1**
Button West: Connect to **A2**
Button East: Connect to **A3**
Button R: Connect to **A4**
Button Start: Connect to **A5**
Button Home: Connect to **A6**

Button UP: Connect to **B0**
Button Down: Connect to **B1**
Button Left: Connect to **B2**
Button Right: Connect to **B3**
Button L: Connect to **B4**
Button Select: Connect to **B5**
Button Share: Connect to **B6**



That's it for that step; the hard part is over!

5: Build the Drum

Important: The SVG files are provided at the correct scale and should NOT be resized. The drum dimensions are precisely calculated to work with the sensor housings and other non-scalable components.

To verify the SVG files are at the correct scale before cutting, check that the 14 mounting holes in the drum base measure exactly 6mm in diameter.

No CNC access? Ask a friend, local shop, or check if your area has a Makerspace. Alternatively, print the SVG files full-size across multiple sheets (ensure your printer is set to 100% scale / "Actual Size"), overlay the paper on your wood as a template, and cut and drill by hand.

B: Prepare the wood	
(📁 B1. Cut all MDF wood pieces per SVG templates in the file packet here.)	You can also cut all of the pieces using a handsaw, jigsaw, etc. If a CNC machine or Laser cutter isn't available. You can check with your local Makerspace for assistance as well
B2. Sand smooth as needed	
Assemble the drum Structure	
B1. Use Wood Glue to glue the two identical rear base plates together (they are 100% identical, just align the holes) clamp securely or weigh down and let dry for several hours.	

B2. Use Wood Glue to glue a Don/Ka top/bottom panel pair together (solid face piece on top + piece with pre-cut holes on bottom). Clamp securely or weigh down and let dry for several hours. Repeat this for the remaining three drum panels.

Ignore the 4 smaller holes in the center, your version will be different and only have 2 holes - this is an example from an earlier version



B3. Use a router or at least sand down the sharp outer, curved edge of the Ka rim faces. This helps prevent stick damage. Do not smooth/route the inside edges at all, just the outside curve.



B4. Using an 8mm diameter drill bit (or the specialized drill bit that came with your threaded wood inserts), locate the 14 pre-cut 6mm holes where the rubber grommets will go, and use those holes as guides to drill - you'll drill into those 6mm holes and turn them into new 8mm wide x 11mm deep holes. Mark your drill bit at the 11mm mark with tape for easy reference so you know when to stop.



B5. Chamfer the rim of the 8mm hole slightly so that the threaded inserts will tighten down flush and feel smooth when your fingers pass over them. This can be done with a sharp screwdriver, knife, etc.



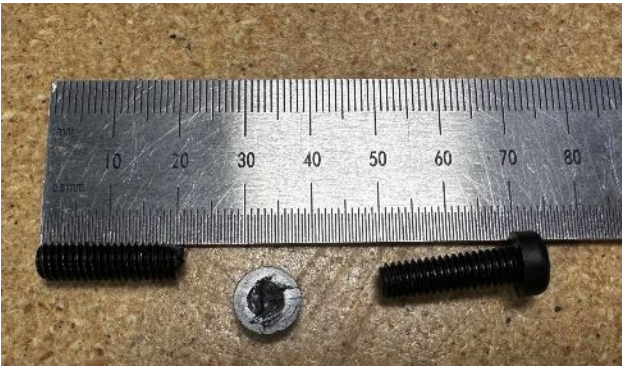
B6. Screw in the 14 individual M6 threaded wood inserts into the appropriate holes until flush (add Superglue to the *outside* of threads to help secure them to wood).



B7. Tap in the 8 individual M3 threaded inserts (2 in each bottom panel) flush to the wood with a mallet (add Superglue to *outside* of threads to help secure to wood) - this is where the sensor housings will mount later on.

Your orientation will be different, this is just an example from an earlier version

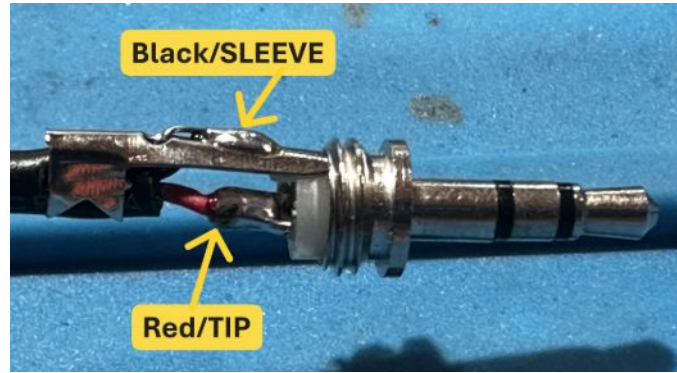


C: Install Rubber Isolators	
C1. Cut the head off of a 20mm M6 nylon bolt	
C2. Apply one drop or less of Loctite to threads on the inside of the rubber isolator threaded hole	
C3. Screw the headless bolt into isolator until it stops	
C4. Add one drop or less of Loctite to inside threads of the M6 threaded inserts on the bottom face of the drum plates	

C5. Screw the nylon bolt of the rubber grommet assembly into the threaded insert in the bottom of the drum faces - finger-tighten until flush. Repeat C1-C5 for the remaining 13 similar holes. This metal bolt will go through the matching holes in the drum baseplate in a later step.	
Note: You will be directed to complete the rest of drum assembly after sensors are installed (end of Section E below)	
D: Print Housings	
D1. Print 4 sets of complete sensor housings ( The files are in the file packet here.)	Use PLA filament. Settings: 0.2mm layer height, 40% Gyroid infill, no supports needed.
E: Assemble Sensors	
 Before you start this step, click here for a video overview of the sensor housing assembly process.	Note: The below steps will need to be repeated 4 times - you will have a total of 4 complete sensor dongles.
( E1. Cut four 12mm neoprene discs per SVG template found in the file packet here.)	
E2. Superglue neoprene disc into the printed housing's bottom center, aligned with the raised ring guide. Press for 30 seconds.	
E3. Take 10" length of Siamese wire, strip both the ends exposing the two wires within.	

E4. Take one end of the wire and solder the two exposed wires to the male TRS jack:

Red connects to the TIP
Black connects to the SLEEVE).



E5. Take the *other* end of the stripped wire, solder red wire to piezo center disc (**near outer edge of center**), and solder black wire to outer brass ring.



E6. Add several drops of Superglue on the surface of neoprene (created in step E2). Center the piezo sensor face up (all-brass side faces *down*) on the neoprene and press together. Hold for 30 seconds. Be sure that wire is laying across the strain relief channel and in the hole cutout.

Pic varies slightly from your version - this was an earlier version. You will have a more pronounced strain relief channel

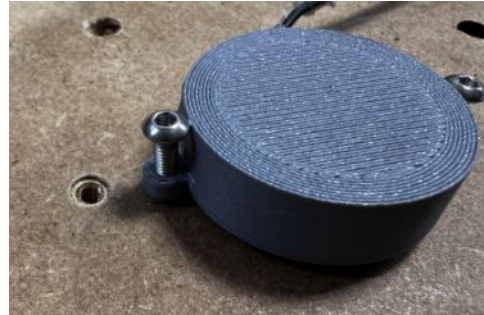


E7. Add a drop of Superglue to the strain relief channel (below wire as well as on topside) and add a few drops to the upper housing around the rim sides, then attach the top and bottom housing pieces, press and hold for 30 seconds.



E8. Mount completed housings to the underside of drum faces using 2 M3x8mm screws for each housing and screw them into the threaded inserts that you added in the earlier step. (be sure that the bottom (side with the neoprene disc inside) of the housing is against the wood).

Tighten snug for even, firm contact with the wood. - but don't over-tighten. Here's a photo of one housing right before it's screwed into the threaded insert.

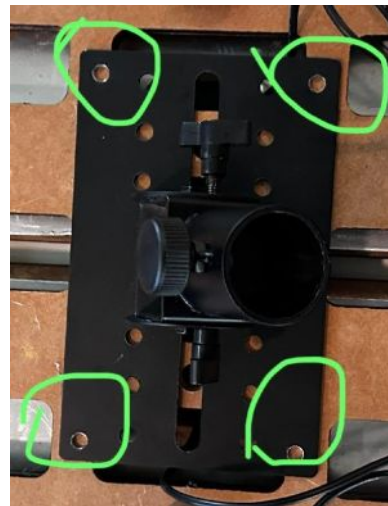


F: Mounting Hardware Assembly

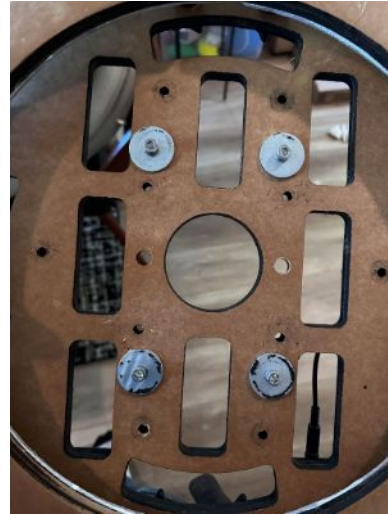
F1. Place your speaker bracket against the backside of your rear base plates so that it is centered and not blocking any other holes. Mark the 4 holes that you will use to mount it with.



F2. Remove the speaker plate and drill your 4 marked holes using a 6mm drill bit.



F3. Feed 4x M6x16 bolts with washers through these mounting holes on baseplate so that they protrude from the rear to attach the speaker mounting plate.



F4. Attach adjustable speaker mount to the speaker bracket, secure with M6 knobs.



F5. Now assemble the rest of the drum structure by feeding the M6x18 bolts on the bottom of the 4 drum faces through base plate holes of the rear baseplate.



F6. Route sensor wires through nearest openings



F7. Secure the drum faces with the M6 knobs on exposed bolts.



F8. 3D Print and Attach TRS barrel mounts with adhesive tape as seen in pictures.

You will not have connected the barrel connectors yet, photo for placement reference.



F9. Connect 3.5mm TRS male drum sensor ends to top of barrel couplers.



6: Control Box Enclosure

You're almost there!

The USB hub and USB DAC steps below are required for the build, but the structural enclosure itself is optional.

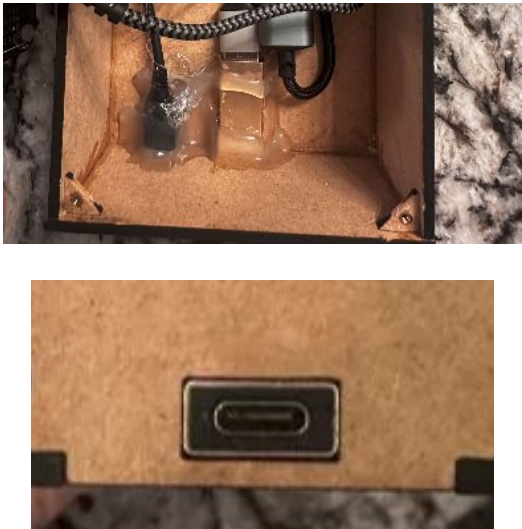
That said, some type of control box enclosure is highly recommended to protect your circuit and components. It keeps parts clean and safe, and allows you to connect and disconnect the drum without exposing the electronics.

Enclosure Options:

- **Custom wood box:** Visit Boxes.py (link [here](#)) to design and create your own
- **Simple alternatives:** Cardboard box or small plastic storage container work fine

I decided against providing plans for the control box since everyone's layout will vary slightly depending on their build choices and creative freedom. The steps below show my personal approach.

G: Connect Sensors To The Circuit	Notes/Photos
G1. Using short male-to-male 3.5mm TRS cables, connect the bottom of the 4 couplers from the last step to the input of the Female TRS jacks on your circuit completed circuit.	Technically that was the last step for the completed controller circuit. The next step will incorporate a few last steps to make the connections more user-friendly, as well as keep your circuit protected in an enclosure.
Final Connections and Housing	Notes/Photos
G2. Mount the 4 Female TRS jacks that are part of your circuit to the wall of your control box so that they face the outside.	 

G3. Connect the USB C to USB A adapter in the wall of your box so that the USB C portion is facing the outside of the box. Use hot glue to affix in place.	
G4. Connect the USB HUB to the USB A side of the adapter you just glued into place.	
G5. Connect the XIAO main controller board to the USB Hub using a short USB C to USB A cable	
G6. Connect your USB DAC to the USB Hub	
G7. Connect the 3.5mm audio out jack on your USB DAC to the short 3.5mm TRS extension cable.	

G8. Connect the 3.5mm Female end of the short audio extension cable you just connected to the wall of your controller box so that it faces outside of the box. Use hot glue to affix in place.



G9. Mount the board with your OLED screen and gamepad buttons so that the screen is visible and the buttons are accessible during gameplay.

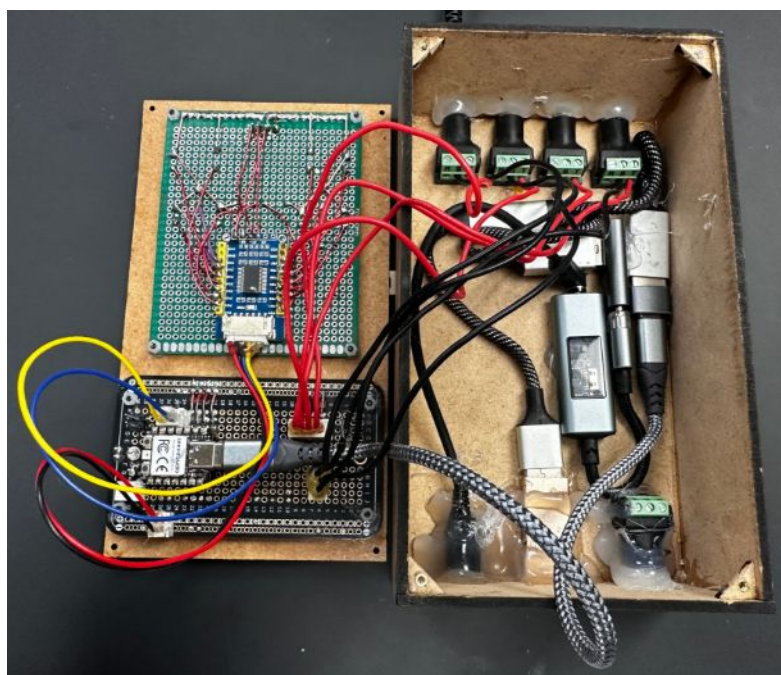


G10. Here's an example of how the inside of your control box may look.

Take your time with wire management. It doesn't have to be the prettiest thing in the world but make sure of the following:

- All connections are secure, and nothing will be bent or stressed or pulled loose when the box is closed.

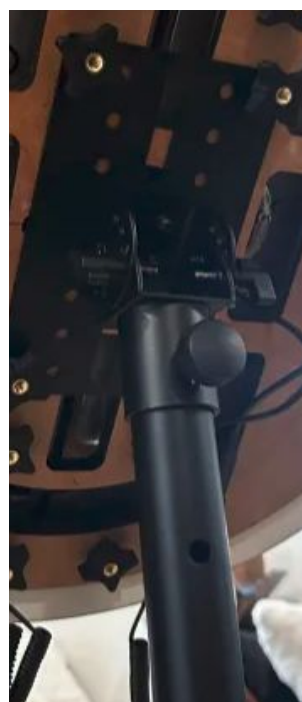
- As much as possible, route your VCC power lines and ground lines away from signal wires and button wires.



H: Floor Stand

Notes/Photos

H1. Mount bracket to the speaker stand



H2. Adjust height/angle for playing comfort

H3. Add your Drum cover, skin, towel, padding, or whichever you have. As my dimensions for the drum in this project are exactly the same as the Arcade Drum, I recommend locating an official Arcade drum skin. One source that seems to always have them in stock is [here](#).

If that isn't an option for you, you can try using a towel, blanket, large mouse pad, thin foam, whatever you want that gives you sound-reducing qualities as well as the amount of bounce that you're looking for.

The beauty of my Project is that you can adjust the Drum and make it perform just how you want regardless of Drum cover.



7: Flash The Firmware

Great job making it this far! Now it's time to prepare the board and flash the firmware. This is the quickest and easiest part of the entire build.

( Download the two flash files in the .zip archive and unzip them. The packet is [here](#).)

Step 1: Enter Boot Mode

Hold down the small "B" button on your actual XIAO RP2040 micro controller board while connecting it to your PC. The board will appear as a removable drive.

Step 2: Wipe the Board

Drag 'universal_flash_nuke.uf2' to the root of the removable drive. This wipes the board for a fresh start and automatically reboots it. The drive will reappear, ready for the next step. Repeat this step one more time for good measure.

Step 3: Flash the Firmware

Drag 'KillerQsOuchiTaikoFirmware.uf2' to the removable drive. After a few seconds, the board will upload the file and automatically reboot. This time, the drive won't reappear—that's correct! Your board is now in controller mode and ready to play.

You're ready for calibration and testing!

8: Calibration & Setup

Configuration:

1. Adjust settings on-the-fly to match your gameplay style and game version. These settings are built into the firmware and can be changed directly from the controller using the OLED display, no coding required.

To access the Main Menu: Hold SELECT button for 1 second.

MAIN MENU STRUCTURE:

1. Controller Modes (12 options)

1. HORI NSW-079 Taiko Drum (Switch)
2. HORI PS4-095 Taiko Drum (PS4, with time limitation)
3. Switch Pro Controller
4. XInput & XInput Analog (Compatible with TaikoArcadeLoader)
5. DualShock 4 (PC/Steam only)
6. DualShock 3
7. Keyboard Player 1
8. Keyboard Player 2
9. Analog Player 1
10. Analog Player 2
11. MIDI
12. Debug

2. Drum Settings

- Drum Thresholds (Left Ka, Left Don, Right Don, Right Ka)
- Big Hit Arcade Mode
 - Detection: On/Off
 - Threshold Level
- SimulTap Mode (Toggle)
- Hold Time (0-255ms)

3. LED Settings

- Brightness (0-255)
- Player Color (PS4 only)
-

4. About

5. Reset

6. USB Flash Mode

NAVIGATION:

- UP/DOWN: Navigate items
- LEFT/RIGHT: Cycle options / adjust values

- EAST (B/Circle): Confirm / Save
- SOUTH (A/Cross): Back / Cancel

Settings persist after power-off when you save before exiting the menu.

Basic Drum Hit Calibration

Complete this before proceeding further.

Long-press SELECT to enter setup mode. Set controller emulation to SWITCH TATACON mode (default). Test all four drum sensors and 14 navigation buttons at the Meloncolle Taiko Controller website (link [here](#)). If automatic drum hits occur without touching anything, don't worry—this is normal during initial setup. The calibration process below will fix it.

CALIBRATION PROCESS:

Goal: No missed notes, no false triggers

THRESHOLD LOGIC:

- Lower = More sensitive (lighter hits trigger)
- Higher = Less sensitive (harder hits needed)

DEFAULT VALUES:

- Left Ka: 70
- Left Don: 55
- Right Don: 55
- Right Ka: 70

PER-PAD CALIBRATION:

1. Hold SELECT → Drum Settings → Drum Thresholds → Select pad
2. If auto-triggering is occurring: Increase threshold by 1-2 until it stops
3. Test light/normal/hard hits—should register on that pad ONLY
4. If other pads trigger (crosstalk): Go to that pad, increase by 5-10
5. Verify: 10 hits at various strengths, all register correctly
6. Alternate between steps 2 and 4 until balanced. This is a one-time setup, but can be adjusted anytime.
7. Repeat the above 6 steps for all 4 pads.

Debounce Delay / Hold Time:

When to change:

- Dropped inputs even when LED confirms hit
- Console needs longer hold time

Adjust: Hold SELECT → Drum Settings → Hold Time
Increase by 5ms if needed. Most platforms work at 25-35ms.

Big Note Mode (Arcade-Style Big Notes)

In arcade Taiko no Tatsujin, Big Notes require hitting the drum harder than regular notes via analog sensing. Home controllers typically require hitting both Don or Ka simultaneously to score full points. I created a way to achieve this with your build! With Big Note Mode enabled, hitting a single Don or Ka harder registers as both pads being hit, giving you full points for Big Notes.

TO ENABLE: Hold SELECT → Drum Settings → Big Hit Mode → Detection: On/Off → Toggle ON

TO SET THRESHOLD:

1. Big Hit Mode → Threshold Level
2. Use Meloncolle website (Switch Tatacon mode)—[link here](#)
3. Hit normally: Single pad lights up
4. Hit HARD: Both pads should light up
5. Adjust: UP = harder hits needed, DOWN = lighter hits work
6. Recommended: 850 (range 700-1200)
7. Adjust until normal hits show as single and HARD hits trigger both drum faces

SimulTap Mode:

Taiko games never require hitting two drums simultaneously (Big Notes don't apply here). By default, this firmware includes mutual exclusivity to prevent crosstalk and double hits by preventing a Don and Ka from being hit/processed at the same time since that situation simply does not exist in Taiko games.

However, for other rhythm games like Clone Hero, you need the ability to hit multiple drums at once. That is why I created SimulTap Mode – which enables this extra feature when needed.

TO TOGGLE: Hold SELECT → Drum Settings → SimulTap Mode → LEFT/RIGHT toggle → EAST save

PS4 Always-On (runs behind the scenes, no user interaction necessary):

This feature is unique to my firmware. DIY controllers that emulate PS4 controllers normally timeout every 8 minutes by design. The existing solution requires extracting proprietary ID keys and manually coding them into the software—a legal gray area that nobody wants to deal with.

I designed a 100% legitimate solution that keeps your controller connected when emulating PS4. No user interaction needed. It's always working—no settings or switches to manage.

Troubleshooting:

DISPLAY:

- Blank OLED: Check I2C (GPIO 6/7), verify 3.3V power
- Frozen: Unplug USB cable and reconnect
- Menu won't open: Hold SELECT for full 1 second
- No animation: Check if sensors register (LED flash)

SENSORS:

- No response: Check diode polarity, TRS wiring
- False triggers: Increase thresholds
- Missed hits: Decrease thresholds, check mounting
- Crosstalk: Increase threshold on triggering pad

MODE SWITCHING:

- Freezes: Unplug/reconnect USB-C
- No change: Press EAST to confirm

AUDIO:

- No sound: Check USB DAC, verify audio output device
- Crackling: Try different USB port
- Latency: Should be <5ms, check game/computer settings

ADVANCED DEBUG: Use Debug Mode to see/adjust raw ADC values (should be 200-400 baseline)

You're Ready to Play!

This controller's 12 emulation modes work with all official versions of Taiko no Tatsujin, as well as clones and open-source versions.

Display Tips While Playing:

- Streak counter resets after 1 second of inactivity
- Character animations confirm every registered hit
- Each character represents its drum position (left to right)
- Consistent missed hits on one pad? Check that pad's threshold

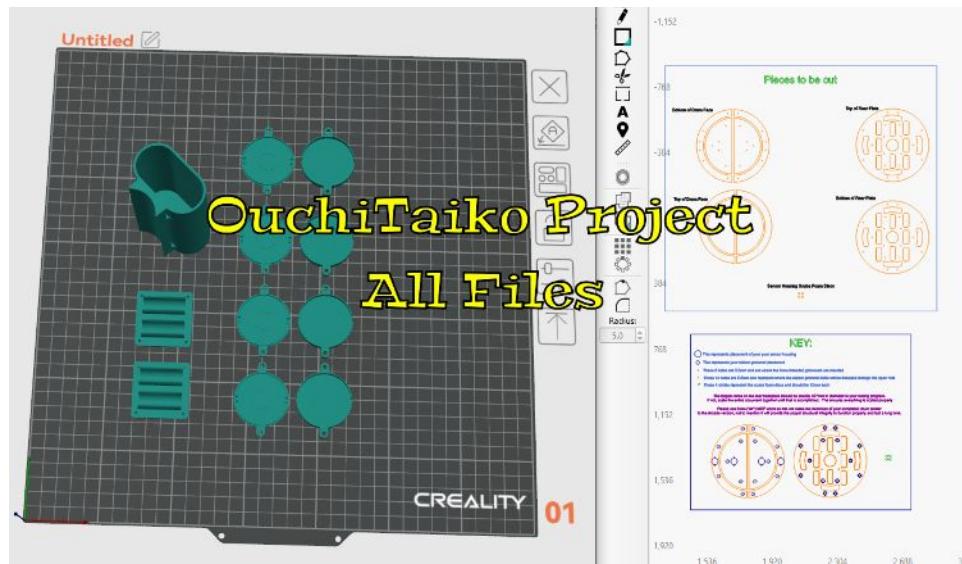
Pro Tips:

- Different drum covers may need threshold adjustment
- Big Hit threshold is typically 2-3x normal threshold
- Use Debug mode to see raw ADC values

If you have trouble with a particular game version, note your current emulation mode and research what others have done to get that controller series working with your game version.

You've just built a professional controller circuit with adaptive baseline technology that automatically maintains perfect calibration—no manual adjustments needed as components wear or age!

9: Files & Downloads



 Here are the files referenced throughout the guide for quick reference:

- To download all SVG and STL Files, click [here](#).
- To Download PDF file of the Circuit Schematic, click [here](#).
- To Download the Firmware File + Nuke File, click [here](#).
- To Download a copy of this guide, use the download link on the landing page at www.OuchiTaiko.com or simply save the PDF using your Web Browser's preferred method.

10: About

This project is a one-of-a-kind Hybrid product that pulls from several amazing resources. As they say, “We stand on the shoulders of Giants...”

The core firmware as well the navigation and OSD hardware portion of the Circuit was adapted from the amazing work by ‘ravinrabbid’ who created the “**DonCon 2040**” Project - link [here](#). You can visit here for more of the nitty-gritty details of the software.

The inspiration for the circuit came from the great, amazingly straightforward circuitry work by ‘kasasiki3’ who created the ‘**HIDtaiko**’ Project (2040 edition) - link [here](#).

Without these two projects, none of this would have been possible.

Credit to ‘Gadgetoid’ on GitHub for his pico-universal-flash-nuke file that helps clean things up between flashes. Their repository can be found [here](#).

Credit to the creative minds over at Boxes.py (link [here](#)) as well for creating the ultimate box-creating resource. It offers endless ideas and guides on how to make enclosure boxes for pretty much anything - especially projects like this.

I want to thank Discord user ‘Allspice.’ I discovered a random message from him in my inbox one day asking if he could help test my Taiko Project when I was finished creating it. He said he had only basic soldering skills, yet he was able to successfully build the entire circuit from scratch as well as the drum! Even more importantly, he helped shape this how-to guide that you are reading right now.

I also want to take a moment to mention a few resources that were invaluable for me during this entire process. These are great if you ever want to go down the rabbit hole of the world of Taiko no Tatsujin modding or Custom Controller creation - with these assets, you have the power to create anything imaginable.

- ‘**Taiko no Tatsujin Modding!**’ Discord Channel - link [here](#).
- ‘**Cons&Stuff :)**’ Discord Channel - link [here](#).
- ‘**OpenStick Community - GP2040-ce Project**’ Discord Channel - link [here](#).

(If you have any questions, please use the Discussion panel on my GitHub here. You can also ask common, non-project-specific questions in the appropriate section of any of the forums mentioned above and tag me if needed.)

Spread The Word:

Please share your results and excitement as well as this guide with your favorite DIY gaming communities. Tag me (KillerQ97) when you do!

Enjoy, Have Fun, and Peace Out!

11: Copyright Information

Copyright & License Transparency

As mentioned above, this project builds upon the outstanding work of the open-source community and strictly adheres to all licensing requirements. **OuchiTaiko** is primarily based on [ravinrabbid's DonCon2040](#), which is licensed under the **MIT License**, and also incorporates elements from [kasasiki3's HIDtaiko](#), licensed under the **Apache License, Version 2.0**.

The most critical requirement of both licenses is to **include a copy of their respective license terms and copyright notices** in all distributions. You can find the complete license documentation here:

- **MIT License (DonCon2040):** <https://github.com/ravinrabbid/DonCon2040/blob/main/LICENSE>
- **Apache License 2.0 (HIDtaiko):** <https://github.com/kasasiki3/HIDtaiko/blob/main/LICENSE>

All custom features, modifications, and improvements documented in this project (including KillerQ's creation of Big Notes detection, SimulTap mode, PS4 always-on, adaptive baseline tracking, critical mode-switching fixes, animated display system, enhanced menu navigation, and any other custom creations or modifications to include code, physical circuitry, Drum, or Drum-related design features and components) are transparently documented and shared under the same open-source spirit. I believe in giving credit where credit is due and maintaining the collaborative nature of the maker community. If you build upon this work, please honor these licenses by including proper attribution and license notices in your derivative projects as well as mentioning and/or tagging me, 'KillerQ.'

Attribution Chain:

- Original DonCon2040 firmware: © ravinrabbid (MIT License)
- HIDtaiko components: © kasasiki3 (Apache License 2.0)
- Custom modifications and features: Created by, and documented in this repository by KillerQ.

For complete legal compliance, any distribution of this firmware (binary or source) must include:

1. A copy of the MIT License from DonCon2040
2. A copy of the Apache License 2.0 from HIDtaiko
3. Copyright notices from both original projects
4. Attribution to all contributors

Full license compliance documentation is maintained in hyperlink form in this guide to ensure transparency and proper credit to all original authors.

Thank you,
KillerQ